

1. USER ANALYSIS: MODELING USER GROUPS

1.1 Why User Analysis Matters

- Designing with users in mind is tricky because humans are multifaceted beings who act differently according to circumstance and context. A person shopping online at home behaves differently than the same person shopping during a work break.
- Traditional user profiles are often too prescriptive—they try to predict exactly what users will do. This approach fails because it doesn't account for the varying contexts that yield different behaviors and decisions.
- The key insight is to focus on understanding the **person** rather than trying to predict their reactions. We should understand their contexts, not just their demographics.

1.2 The Matrix Technique

The Matrix Technique is a structured brainstorming tool that helps make sense of user patterns by exploring what we know and don't know about users through the lens of their behavioral contexts.

What the Matrix Technique helps you do:

- Establish who your key audiences are and understand what makes them different from each other
- Understand or validate value propositions—what value does your product offer to different user types?
- Prioritize research areas by identifying where you have solid data versus where you're making assumptions
- Pinpoint which features are vital for which user groups
- Unpack hidden assumptions that team members and stakeholders make about users

1.3 The Seven Steps of Matrix Technique

STEP 1: Draw Your Matrix

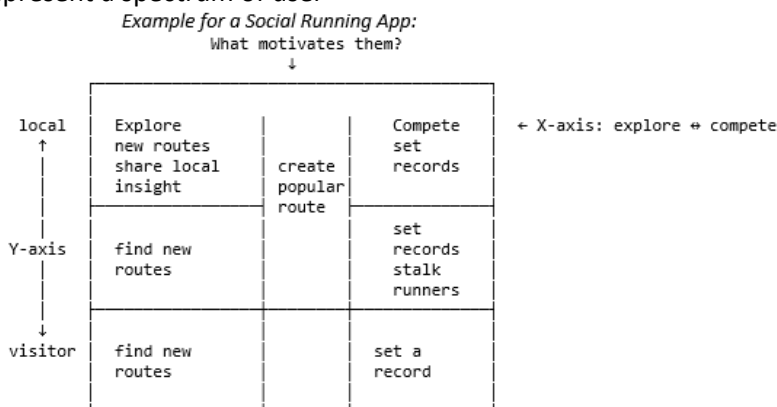
Create a 3x3 grid. This size provides enough variation to capture meaningful differences without becoming overly complex or requiring precision you don't yet have.

STEP 2: Identify Important Axes

Think about how many different "selves" of your users can exist in the system you're designing. Identify behavioral attributes as opposing pairs that represent a spectrum of user behaviors.

Common behavioral axis pairs include:

- **Explorative ↔ Competitive:** "I want to explore" vs "I like to compete/race"
- **Frequent ↔ Occasional:** "I use this several times a week" vs "I use this rarely"
- **Visitor ↔ Local:** "I'm new here" vs "I know this place well"
- **Social ↔ Individual:** "I want to connect with others" vs "I prefer working alone"
- **Curious ↔ Engaged:** "I'm just browsing" vs "I'm committed to this"



STEP 3: Identify Key Questions

For each cell in your matrix, ask questions that help you understand that specific user type:

- "What would users in this cell want to do?"
- "How would they feel about this feature?"
- "What problems would they encounter?"
- "What would motivate them to use the product?"
- "What would frustrate them?"

STEP 4: Fill the Matrix

Fill in each cell based on your insights. As you do this, be honest about what you actually know versus what you believe or assume. Mark cells differently:

- **Known insights:** Things you have evidence for (from research, data, interviews)
- **Assumed insights:** Things you believe but haven't validated
- **Unknown:** Cells where you simply don't know the answer

This distinction is crucial because it shows you where you need to do more research.

STEP 5: Iterate with Different Questions

Create new grids with different questions or even different axes. Each new matrix reveals different aspects of your users. For example:

- First matrix: "What can they do?" (features)
- Second matrix: "How do they feel?" (emotions)
- Third matrix: "What problems do they have?" (pain points)

STEP 6: Analysis and Key Insights

Look across all your matrices for patterns. You might notice that:

- Certain user types appear consistently across multiple matrices
- Some cells remain empty or unknown—these are research priorities
- Certain features matter to multiple user types while others are niche
- There are clusters of related behaviors

STEP 7: Taking It Further

Use your matrix insights to generate:

- **User Personas:** Detailed fictional characters representing each key user type
- **Mental Models:** How different users think about the problem space
- **User Journeys:** Step-by-step paths through your product for each user type
- **Product Roadmap:** Prioritized features based on which user types you want to serve first

2. USER PERSONAS

2.1 What Is a Persona?

A persona is a way to model, summarize, and communicate research about people who have been observed or researched. It's depicted as a specific person but is not a real individual—rather, it's synthesized from observations of many people.

Key characteristics of personas:

- Each persona represents a significant portion of people in the real world
- Personas enable designers to focus on a manageable and memorable cast of characters instead of trying to design for thousands of individuals
- They help designers create different designs for different kinds of people
- Most importantly, personas help you design for a **specific somebody** rather than a **generic everybody** (designing for everybody often means designing for nobody)

2.2 How Personas Are Created: The Eight-Step Process

STEP 1: Pick a Representative Unit of Users

Decide how many interviews you need to conduct or survey responses you need to collect to be confident in your conclusions. This depends on:

- How many different user types you expect to have
- How much variation exists within your user population
- Your available resources and time constraints

STEP 2: Be Creative to Get Data Points

Gather open-ended responses using multiple methods:

- **Surveys:** Online questionnaires that capture written responses
- **Virtual/In-person Discussion Groups:** Focus groups where users discuss their experiences
- **Reverse Q&A Sessions:** Let users ask questions—their questions reveal their concerns and priorities
- **Contextual Interviews:** Interview users while they perform tasks
- **Diary Studies:** Have users record their experiences over time

The key is to get rich, qualitative data about users' goals, challenges, behaviors, and preferences.

STEP 3: First Pass at Personas

Read through all your responses carefully and systematically:

- Find a quiet spot where you can focus without interruption
- Jot down common words and phrases as you go
- Note patterns in goals, frustrations, behaviors, and preferences
- Group respondents based on the commonalities between them
- Look for natural clusters of similar users

STEP 4: Fill in Your Template

Using your groupings from Step 3, create unified persona profiles:

- Formulate group responses into single, unified descriptions
- You should be able to confidently convey main challenges, needs, and goals based on the group's responses

- Keep the total number of personas between **3 to 6**—fewer than 3 means you're oversimplifying, more than 6 becomes hard to remember and use

STEP 5: Validate Through User Interviews

Conduct one-on-one user interviews to validate your conclusions:

- Talk to real users who fit each persona profile
- Verify that your persona descriptions accurately reflect real people
- Refine based on what you learn
- Add details you might have missed

STEP 6: Categorize Users into Personas

Place each user you've studied into a persona based on your criteria:

- Users should fit naturally into one persona
- If a user fits multiple personas equally, your personas might need refinement
- Track how many users fall into each persona to understand relative importance

STEP 7 (Bonus): Note Unimportant Users

Keep track of users who don't fit any of your personas:

- If too many users don't fit, you may need additional personas
- Or your existing personas may need adjustment
- This is also valuable for knowing who you're NOT designing for

STEP 8 (Bonus): Name Your Personas

Give each persona a memorable name that reflects their characteristics:

- Examples: "Marketer Mary," "Developer Dave," "Busy Parent Paula"
- Names make personas easier to remember and reference in team discussions
- Names humanize the data and make personas feel like real people

2.3 Persona Template Components

A complete persona typically includes:

- **Name and Photo:** A fictional name and stock photo to humanize the persona
- **Demographics:** Age, occupation, education, location (relevant details only)
- **Goals:** What they want to accomplish
- **Frustrations/Pain Points:** What prevents them from achieving their goals
- **Behaviors:** How they currently approach tasks
- **Technology Comfort:** Their relationship with technology
- **Quote:** A characteristic statement that captures their mindset
- **Scenario:** A brief story showing them in context

5 Heuristic Evaluation

Heuristic evaluation is a type of predictive evaluation where experts evaluate a user interface against a set of usability principles (heuristics).

Common Usability Heuristics (Nielsen's 10 Heuristics):

1. **Visibility of system status:** The system should always keep users informed about what is going on through appropriate feedback within reasonable time.
2. **Match between system and real world:** The system should speak the users' language, with words, phrases, and concepts familiar to the user rather than system-oriented terms.
3. **User control and freedom:** Users often choose system functions by mistake and need a clearly marked "emergency exit" to leave the unwanted state.
4. **Consistency and standards:** Users should not have to wonder whether different words, situations, or actions mean the same thing.
5. **Error prevention:** Even better than good error messages is a careful design that prevents problems from occurring in the first place.
6. **Recognition rather than recall:** Minimize the user's memory load by making objects, actions, and options visible.
7. **Flexibility and efficiency of use:** Accelerators—unseen by the novice user—may speed up interaction for expert users.
8. **Aesthetic and minimalist design:** Dialogues should not contain information that is irrelevant or rarely needed.
9. **Help users recognize, diagnose, and recover from errors:** Error messages should be expressed in plain language, precisely indicate the problem, and constructively suggest a solution.
10. **Help and documentation:** Even though it's better if the system can be used without documentation, it may be necessary to provide help and documentation.

6. ETHNOGRAPHIC TECHNIQUES FOR HCI EVALUATION

6.1 Understanding Observation

Observation involves watching and listening to users. Even casual observation of users interacting with software can tell you an enormous amount about:

- What they do (actual behaviors, not just what they say they do)
- The context in which they do it (environment, tools, social factors)
- How well technology supports them (where it helps, where it fails)
- What other support is needed (training, documentation, better design)

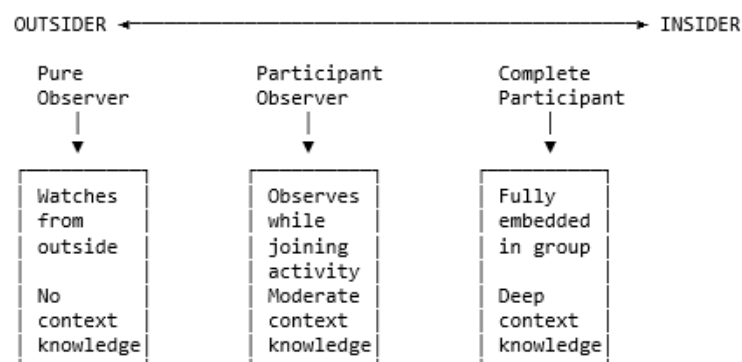
6.2 Observation Settings

Controlled Laboratory Conditions (Usability Testing):

- Structured environment designed for observation
- Video cameras, one-way mirrors, logging software
- Tasks are predefined
- Variables are controlled
- Results are quantifiable

Natural Environments (Field Studies):

- Observation happens where users actually work or live
- Real tasks, real contexts, real interruptions
- More ecological validity (reflects real-world use)
- Harder to control variables
- Richer, more nuanced insights



6.3 The Outsider-Insider Spectrum

The degree of immersion that evaluators adopt varies across a broad outsider-insider spectrum. Where a particular study falls along this spectrum depends on its goals and on practical and ethical constraints.

Colin Robson's Four Levels of Participation:

1. **Complete participants:** Full insiders who are part of the group being studied. They have deep knowledge but may lose objectivity.
2. **More marginal participants:** Involved with the group but maintain some distance. Can participate while still observing.
3. **Observers who also participate:** Primarily observers who occasionally join activities to better understand them.
4. **People who observe from outside and do not participate:** Pure observers with complete objectivity but limited contextual understanding.

6.4 Types of Observation

Quick and Dirty Observation

- Can occur anywhere, anytime with minimal planning
- Evaluators go into a school, home, or office to watch and talk to users casually
- Get immediate feedback about a prototype or product
- Evaluators can also join a group for a short time (slightly more insider role)
- Ways of finding out what's happening quickly and with little formality

Observation in Usability Testing

- Video and interaction logs capture everything the user does
- Keystrokes, mouse clicks, and conversations are recorded
- Observers watch through one-way mirrors or remote TV screens
- Data is used to analyze what users do and how much time they spend
- Provides insights into **affective reactions** (sighs, frowns, scowls indicate frustration)
- Highly structured with predefined tasks and metrics

Observation in Field Studies

- Observers may be anywhere along the outsider-insider spectrum
- Philosophy and practices influence how data is collected, analyzed, and reported
- More naturalistic and less controlled than lab studies
- Can reveal unexpected insights that structured testing would miss

Participant Observation

- Evaluators participate with users to learn what they do, how they do it, and why
- Must share experiences and learn social conventions:
 - Beliefs held by the group
 - Protocols and procedures they follow
 - Dress codes and appearance norms
 - Communication styles
 - Language and terminology
 - Non-verbal cues
- As Fetterman (1998) described: "Participation in the lives of the people under study with maintenance of a professional distance"

Ethnography

- Some view ethnography as synonymous with participant observation
- Others view participant observation as just one technique used in ethnography
- True ethnographic studies typically take weeks, months, or even years
- In interaction design, shorter ethnographic-style studies are more common due to time and budget constraints
- Provides deep, contextual understanding of user behavior

6.5 Case Study Comparison: Field vs. Lab

Scenario 1: Field Study (WAP Phone Evaluation)

A usability consultant joins a group testing WAP phones during a visit to Washington, DC:

- The group needed to find a restaurant (real need)
- They used the WAP phone to find restaurants within five miles (real task)
- They found telephone numbers, called to ask about menus, made a booking (real outcome)
- The consultant observed button size problems and screen size issues
- Discussion confirmed initial impressions

Advantages:

- See the real situation with real problems
- Experience the delight or frustration users actually feel
- Context reveals things lab testing would miss

Disadvantages:

- Observer may not be fully objective
- Data is qualitative and hard to compare
- Results depend on specific circumstances

Scenario 2: Lab Study (WAP Phone Evaluation)

Predetermined task in a usability laboratory:

- Task: Find the telephone number of a specific restaurant
- Controlled conditions, measured outcomes
- Same task for all participants

Advantages:

- Can compare users directly
- Can calculate averages and statistics
- More objective measurements

Disadvantages:

- Artificial situation doesn't reflect real use
- Misses contextual factors
- Tells nothing about real-world motivations

Key Insight: Both approaches have merits. Which is better depends entirely on the study's goals. Often, the best approach combines both—lab testing for quantitative metrics, field studies for contextual understanding.

7.1 The User Behavior Formula

Applied psychology ensures reliable user-computer interaction. The psychology of an individual user centers on cognitive abilities like learning, performing, and reasoning.

The Core Formula: USER BEHAVIOR = Users' Goals + Task Structure + Users' Knowledge + Users' Processing Limits

Understanding each component:

- **Users' Goals:** What objectives users want to accomplish. Goals drive behavior—users don't interact with systems for fun but to achieve something.
- **Task Structure:** The steps users would "like" or "not like" to perform. The user interface is one of the most important parts of task structure because it determines what steps are necessary.
- **Users' Knowledge:** What users know about the existing task structure. This includes both procedural knowledge (how to do things) and conceptual knowledge (understanding why).
- **Users' Processing Limits:** Mental processing constraints, specifically:
 - **Restricted short-term memory:** Humans can only hold about 7 ± 2 items in working memory at once
 - **Propensity to make errors irregularly:** Humans make mistakes unpredictably, even on tasks they've done correctly before

7.2 Controlling User Behavior (Improvement Strategies)

Since we understand the components of user behavior, we can improve systems by addressing each:

Strategy 1: Improve Task Structure

- Make improvements in the user interface
- Reduce the number of steps required
- Make each step clearer and easier
- Provide better feedback at each stage

Strategy 2: Increase Users' Knowledge

- Teach users more efficient methods
- Provide good documentation and tutorials
- Design interfaces that teach as users use them
- Make the conceptual model clear

Strategy 3: Control Error Propensity

- Implement efficient error recovery (undo, confirmation dialogs)
- Prevent errors before they happen (constraints, defaults)
- Make errors reversible where possible
- Provide clear error messages when errors occur

Strategy 4: Organize User Goals

- Automate tasks that could be performed interactively
- Break complex goals into manageable sub-goals
- Align system capabilities with user goals

Key Insight for Processing Limits:

Break larger tasks into smaller ones. This provides:

- **Less memory load:** Users only need to remember the current step
- **Bounded scope of errors:** Mistakes affect only the current subtask, not the entire process

7.3 Dimensions of User Behavior

Trade-offs Between Dimensions:

These dimensions often involve trade-offs. For example:

- **Speed vs. Learning:** A faster interface for experts might be harder to learn for novices
- **Functionality vs. Simplicity:** More features mean more complexity
- **Quality vs. Time:** Better output might require more time

Designers must decide which dimensions to prioritize based on their users' needs.

7.4 User Classification by Experience

Novice Users:

- In the initial stage of learning
- Everything is a new problem to solve
- Primary concern: Learning how to perform tasks
- Least concerned with time/efficiency
- Need clear guidance, tutorials, and forgiving error handling
- Benefit from: step-by-step instructions, limited options, helpful defaults

User behavior can be measured along multiple dimensions:

DIMENSIONS OF USER BEHAVIOR	
FUNCTIONALITY	The series of tasks that can be performed What capabilities does the system offer?
LEARNING	Time taken to learn given tasks How long before users become proficient?
TIME	Time taken to perform given tasks How efficient is task completion?
ERRORS	Number and severity of errors How often do mistakes happen? How bad?
QUALITY	Excellence of output How good are the results users achieve?
ROBUSTNESS	Comfort in performing new/unexpected tasks Can users handle novel situations?
ACCEPTABILITY	Overall subjective feeling about system Do users like using it?

Expert Users:

- Have mastered a portion of the system
- Perform tasks with routine cognitive skill (almost automatically)
- Primary concern: Speed and accuracy
- Are immune to minor variations in interface structure
- Want: shortcuts, customization, minimal interference
- Frustrated by: unnecessary confirmations, forced wizards, hidden advanced features

7.5 The User Interface: Two Views

The Designer's View:

- Terminal comprising hardware and software
- Collects, interprets, and displays data
- Technical focus on inputs and outputs

The Psychological View:

"The user interface is any part of the computer system that the user comes in contact with—either physically, perceptually, or conceptually."

This psychological view has important implications:

- Users create a **conceptual view** of the system from its entire behavior, not just the screen
- The user interface includes **training and documentation**, not just the software itself
- Interface design must be considered **early in the design process**—it cannot be effectively added as an afterthought

7.6 Conceptual Models

A conceptual model is information that classifies the overall working of a system and its use. It must derive from the overall behavior of the system, and users must be aware of the conceptual model being followed.

Why Conceptual Models Matter:

- Users form mental models of how systems work
- If the actual system matches their mental model, the system feels intuitive
- If there's a mismatch, users become confused and make errors
- Good design makes the conceptual model clear and consistent

8. USER ACCEPTANCE OF INFORMATION TECHNOLOGY (TAM)

8.1 The Technology Acceptance Model

The Technology Acceptance Model (TAM) specifies user behavior and provides an overall view of design choices that impact user acceptance. It helps predict and evaluate user acceptance of information systems.

The Critical Finding:

"Perceived usefulness" was 50% MORE influential than "ease of use" in determining system usage.

This means that people will tolerate a difficult interface if the system helps them accomplish their goals. However, an easy-to-use system that doesn't help them do their job will be rejected.

8.2 Key Definitions

Perceived Usefulness:

"The degree to which an individual believes that using a particular system would enhance his or her job performance."

This is about results. Does the system help users accomplish what they need to accomplish? Does it make them more effective?

Perceived Ease of Use:

"The degree to which an individual believes that using a particular system would be free of physical and mental effort."

This is about the experience. How much effort does it take to use the system? Is it intuitive or frustrating?

8.3 The TAM Relationships

Key Relationships:

1. **Perceived ease of use directly impacts perceived usefulness** (but not vice versa)
 - When something is easier to use, people find it MORE useful

- An easier interface means less effort spent on the tool, more on the task

2. System design features indirectly impact attitude and usage

- Design features don't directly make people use a system
- They work through perceived usefulness and ease of use

3. Both perceptions contribute to attitude

- Attitude toward using the system combines usefulness and ease of use
- Attitude then drives actual usage behavior

8.4 Design Implications

To Make a System More Useful:

1. **Add new functional features:** Give users capabilities they need
 - Understand what users are trying to accomplish
 - Provide tools that help them do it better
2. **OR make existing functions easier to use:** Reduce effort required
 - Simplify workflows
 - Reduce steps
 - Improve feedback

The Critical Insight:

"Users could use a difficult interface to accomplish their tasks, but an easier-to-use interface that does not help them perform their job could NOT be accepted."

This means:

- Usefulness trumps usability when push comes to shove
- But usability contributes to perceived usefulness
- The best systems are BOTH useful AND easy to use

8.5 The Key Factor in User Acceptance

The most significant barrier to user acceptance is **lack of user friendliness**. This can be overcome by:

- Adding user interfaces that increase usability
- Designing with users' mental models in mind
- Testing with real users throughout development
- Iterating based on feedback

9. UNIVERSAL USABILITY

9.1 What Is Universal Usability?

Universal usability means achieving a state where **90% of household people are using computers at least once a week**. This ambitious goal requires addressing the needs of diverse users with different abilities, knowledge, and circumstances.

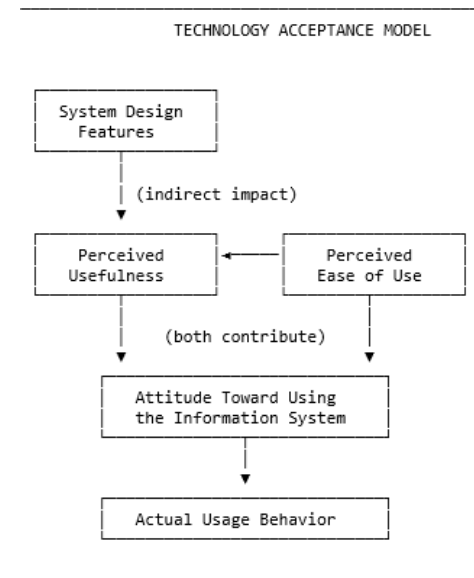
9.2 The Current Problem

Key Statistics:

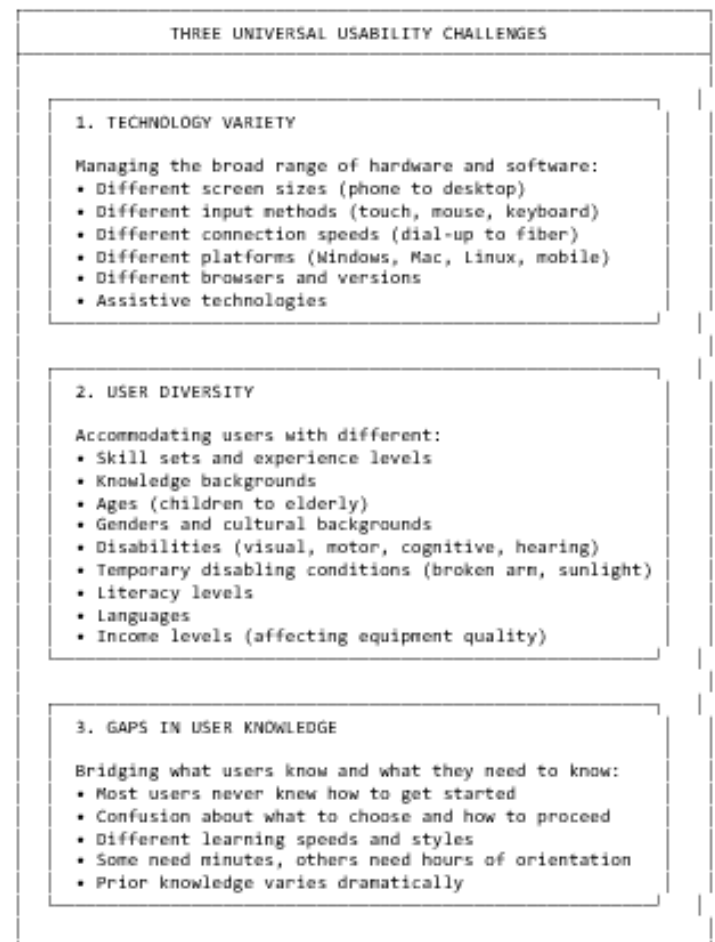
- On average, **5.1 hours per week** is wasted in front of computers due to usability problems
- More time is wasted trying to use computers than to use highways
- This raises user frustration and anxiety
- A large number of potential users remain non-users because systems are too difficult

The Challenge:

- Lowering the cost of computers can raise the number of users
- But there's also a critical need to improve interface and information designs
- Both infrastructure (access) and usability (design) must improve together



9.3 The Three Universal Usability Challenges



9.4 Application Example: E-Government Services

Governments are trying to serve large numbers of citizens through digital services (like electronic tax filing). This presents unique challenges:

- Citizens have vastly different backgrounds and skill sets
- Cannot exclude any citizen based on technical ability
- Must understand problems of both **users** (those who try and struggle) and **non-users** (those who don't try at all)
- High stakes—citizens depend on these services

9.5 Counter-Argument to Universal Usability Criticism

The Criticism:

"Catering to low-end technology, disabled citizens, and non-skilled users results in systems that are least beneficial for most users."

The Response:

When you design for a broader spectrum of users, you create a wider range of design solutions. This leads to innovations that benefit ALL types of users, not just the edge cases.

Examples:

- Curb cuts (sidewalk ramps) help wheelchairs, but also strollers, carts, and bicycles
- Closed captions help deaf users, but also people in noisy environments or learning the language
- Voice control helps blind users, but also drivers and people with full hands

9.6 Strategies for Universal Usability

1. Layered Approaches

Create multiple versions for different user segments:

- Universities segment websites for: prospective students, current students, alumni, faculty
- Banks segment for: personal banking, business banking, wealth management
- Software provides: beginner mode, advanced mode

Segmentation tools help users self-identify and reach appropriate content.

2. Accessibility Features by User Type

For users with **poor reading skills**:

- Provide simpler text versions
- Offer multiple language options
- Use clear, plain language
- Include visual aids and icons

For **partially sighted or elderly** users:

- Allow increasing font size
- Provide high contrast options
- Note: These options are rarely available in control panels, help messages, and dialog boxes—a design gap

For **blind** users:

- Create customized interfaces with voice support (screen readers)
- Ensure proper semantic HTML structure
- Provide text alternatives for images
- Support keyboard-only navigation

For **expert** users:

- Offer customization options
- Provide macros for repeated operations
- Support special-purpose devices and shortcuts
- Allow interface personalization

3. Bridging Knowledge Gaps by User Level

KNOWLEDGE GAP SOLUTIONS BY USER LEVEL	
NOVICE USERS	• Simple, step-by-step instructions • System overviews that explain concepts • Error prevention (constraints, warnings) • Effective tutorials with practice • Forgiving interfaces (easy undo)
INTERMITTENT USERS	• Constructive, context-sensitive help • Reminders of how things work • Consistent interfaces (less to remember) • Clear navigation (easy to find features)
EXPERT USERS	• Compact, dense presentations • Keyboard shortcuts and accelerators • Customizable interfaces • Minimal interruptions
ALL USERS	• Online help via email • Telephone support • Video conferencing assistance • Shared screen support • Community forums